

Economic Environment Surveillance to predict Issues and Disruptions through Multimodal Deep Learning, Smart Education, and Secure Information

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Abstract

This study examines the economic environment surveillance to predict issues and disruptions through multimodal deep learning, smart education, and secure information. It is known that education plays a leading role in raising awareness, mitigating problems and perturbations, and general human security, to improve economic activities, in sustainability. It relies on a CNN-LSTM-based multimodal deep learning model. The research discusses integrating artificial intelligence (AI), Internet of Things (IoT) technologies, big data, smart data, supercomputing, and social networks into economic surveillance and preventing issues and perturbations using smart education and safe news. This multimodal model improves economic activity by providing interactive smart education for warning people, especially managers about an ongoing event to make quick decisions. LSTM's long term dependency learning, using commercial, marketing, or financial data as input, and encapsulation, using connected objects, smart data, and social networks, will enable you to identify relevant and secure information. Our new model is shown to learn to locate and recognize multiple aspects of marketing, business, and finance. The model is evaluated on the challenging tasks of electricity fraud detection in business and finance. It is shown that it is more accurate than the state-of-the-art of other neural networks and uses less training time.

Keywords: Big Data, CNN-LSTM-based multimodal deep learning model, Connected objects (sensors), Economic activities, Smart education, Secure information, Social networks

1. Introduction

In everyday life, various business, marketing and financial information circulates from multimodal sources (connected objects, smart data and multiple social media sources), some of them on all social networking channels. Capturing content [2] from connected objects, smart data or multiple social media sources is important for a comprehensive view of the event. It is a significant challenge for managers to retrieve the vast amount of information shared across the web. To help managers identify and filter relevant, useful shared content, several artificial intelligence systems [1] & [2] are designed to monitor the network environment. Some work focuses solely on the use of social networks as a source of information, while others are dedicated to warning [1], education [2] and awareness raising [3, 4]. Designing multimodal deep learning to enhance economic activity using big data, high-performance computing, and secure information education. A new model based on CNN-LSTM for learning long-term dependencies is proposed. It proposes a new model for improving economic activities, based on a CNN-LSTM for learning long-term dependencies. This new approach enables the integration of artificial intelligence technologies, deep learning, and social media into multimodal deep learning for the management of economic activities [6, 7].

Building on previous work [2, 3, 5, 6, 7], which forms the background for this approach, this new model combines representation training with warning and smart education, while incorporating encapsulations from connected objects, smart data and multiple social media sources. This research study involves monitoring the network environment to identify content relevant to an impending economic crisis. This could be a public health crisis. Once the data is retrieved and cleansed of non-informative data, it is used to update the data (alert or intelligent information) for citizens and managers who need to make quick and effective decisions to modify trends to prevent large-scale, uncontrollable crises in the community (lack of flour, milk, especially for children, drinking water...) which could trigger demonstrations, loot, sabotage, etc.

2. Background & Related Works

There is a lot of research on forecasting of financial time series [8, 9]. Research on fraudfinding is common [8]. In contrast, there are only two studies on pricing [10] and only one on product placement in supermarkets. Twitter (now called X) is the first social network used in news tweeting applications [11, 12, 13]. It is followed by the NN-based crisis management model to improve alerting [1], which uses Twitter (now called X) & Facebook. The FFNN-based ALE follows to improve warning and education from the web: It is the first to use multiple sources with the whole web.

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Electricity fraud detection approach [14] using a combination of Long Short-Term Memory (LSTM) methods and robust Exponential and Holt-Winters Smoothing (EHWS) methods, using sensors to improve the accuracy and efficiency of fraud detection mechanisms, while studying the fraudulent behavior of electricity consumers. It addresses the huge problem of non-technical losses (NTL), also known as electricity fraud, for electricity distribution utilities. A total of 385,714 fake goods [15] will be confiscated by the Algerian tax authorities in 2020, according to the anti-fraud department of the Algerian tax administration ¹.

There are some other works, namely LSTM [8], RNN [3], and Deep CNN-LSTM model [5, 6, 7] to improve warning, awareness, and education in crisis events using the whole web. There are other models to improve preventing fake news, namely improving commerce and business, financial time series forecasting, fraud detection, and marketing using deep learning, big data, and high-performance computing while avoiding fake news [5, 6, 7], followed by deep learning-based crisis management with intelligent education in environmental sustainability while avoiding false news [4], followed by context-based false news detection [16], and fake news detection using CNN-BLSTM based hybrid model through context social [4]: All these models use the whole Web. There is the first model that uses smart data & the whole web: it is the RNN-based ALE [3] that improves awareness. Other applications based on sensor technology [17, 18] are being used as the IoT advances. Thus, the study could integrate multimodal deep learning. The proposed model includes multimodal deep learning design and implementation to enhance economic activities using big data, high-performance computing, and education.

If the content is processed quickly and efficiently, online content contains important information that can contribute to rapid decision-making in support of affected communities. Several types of processing techniques, ranging from comparable document-based data, statistical analysis, and natural language processing to machine learning [1, 2] and computational linguistics, are developed for different purposes without fully utilizing these data, despite the existence of certain resources such as tagged data and standardized lexical resources.

The study proposes a new crisis management model that focuses on multimodal deep learning for warning and smart education, building on previous work [3] to improve fraud detection and forecasting of financial time series. Networking is an essential part of the online activities of website users. Recent developments in networking use reflect the fact that not only is the number of users growing, but also the number of applications (see Table 1).

While the predominant function of networking continues to be social interaction, networking sites are also considered to be the fourth most popular source of information. Networking before, during, and after an event can facilitate data sharing. Thanks to Web 2.0, the spread of social media is transforming

knowledge from expert knowledge to everyday knowledge co-produced by different stakeholders. In recent years, a growing number of studies have explored the use of data from social networks to gain knowledge in areas of human activity as diverse as the detection of diseases as epidemics and the prediction of the stock market. However, it is a challenging task to make sense of this voluminous and fast-moving data.

Table 1: Comparing techniques and methods in models using various Sources of Capturing Content

Ref	Method	Sources of Capturing Content
[12]	Game-based learning in flood crises	Twitter
[13]	An educational objective of higher education, specifically related to	Twitter
[19]	Socio-temporal context summary	Twitter
[20]	Semi-automated AI-based classifier for crisis response	Twitter
[11]	Abstract-extractive approach to summarising contextual tweets in a crisis situation	Twitter
[1]	NN-based warning model	Facebook & Twitter
[2]	FFNN-based ALE to enhance warning and education	All the Web
[8]	LSTM-based ALE to enhance warning and education	All the Web
[3, 4, 16, 5, 6, 7]	CNN-LSTM-based Managing Crisis Model to enhance Warning, Awareness & Education via the Web	All the Web
[21]	RNN-based model	Smart Data & All the Web
[17, 18]	Sensor-based model	Sensor
[14]	Detection of economic electricity meter anomalies and customer fraud cases	Sensor
Our New Approach	Traffic environment monitoring for warning, awareness and sustainable with safety education	Connected objects (Sensors), smart data & All the Web

Concerning the online methodology, the listening and monitoring approach of this study consists in the collection of all the content issued from social media conversations, news or any information on the web; the cleaning of the data from duplication and replication content, enabling relevance thanks to neural learning, with the learning corpus obtained thanks to the tagged messages. Verification and analysis of the relevant content is carried out by the emergency management model. This ensures adequacy and alerts managers.

The shelves on which products are displayed are one of the most important resources in the retail environment, given the wide variety of products and shopping behaviors. Retailers can increase profits and reduce costs through proper management of shelf allocation and product display. Through the use of learning models in the organization of shelves in supermarkets by grouping products that are usually bought together, the research study can extract the following relationship: customers who buy product X at the end of the week, during the summer, usually also buy product Y (see Table 2). To improve the

¹<https://www.aps.dz/en/economy/39561-customs-over-385-000-counterfeit-goods-seized-in-2020>

Table 2: Economic activity examples

Economic activities	Models
Novelty fraud detection	[22]
Trading performance	[23]
Forecasting the exchange rate	[10]
Trading on the stock exchange	[9]
Improving warning and education with RNN-based Model	[21]
Improving warning and education with LSTM-based ALE	[8]
Improving warning and education with CNN-LSTM-based Model	[5, 6]
Financial time series forecasting	[24]
Company shares	[25]
Detection of economic electricity meter anomalies and customer fraud cases	[14]
385,714 counterfeit products were confiscated by the Algerian tax authorities in 2020, according to the anti-fraud service of the Algerian tax administration ²	[15]

accuracy and efficiency of fraud detection mechanisms and to study the electricity consumers fraud behavior, [14] combines LSTM (Long Short-Term Memory) and EHWS techniques. It addresses the huge problem of non-technical losses (NTL), also known as electricity frauds, for electricity distribution companies. A total of 385,714 counterfeit goods will be seized by the Algerian tax authorities in 2020, according to the anti-fraud service of the General Directorate of the Algerian tax authorities³.

Another example is the credit institution, which uses data mining to decide whether or not to grant credit based on the applicant's profile, their application, and their previous credit history. There is also overbooking (optimizing the number of seats in planes, hotels, etc.), targeted offers (organizing advertising campaigns, and promotions), and analysis of business practices, strategies, and their impact on sales. This knowledge is initially unknown. It may be correlations, patterns, or general trends in the data. Experimental data is needed to correct the system or to estimate some difficult mathematical modeling parameters. Data Mining is a field that has emerged with the explosion of the amount of information stored, with significant progress in particular in processing speeds and storage media. Data Mining aims to find valuable information in large amounts of data that can help understand the data or help predict how future data behaves. Since its inception, data mining has used various statistical and artificial intelligence tools. It is a key component of big data, big data analytics, and, more recently, intelligent streaming. It is often defined as the process of discovering new knowledge by examining large amounts of data (stored in warehouses or streaming) using pattern recognition, and statistical and mathematical techniques. Fraud detection is interesting because it can be formulated as unsupervised and/or supervised classification (see Table 3).

In the unsupervised learning category, class labels are either unknown or assumed to be unknown, and clustering techniques

are used to find (i) distinct clusters containing fraudulent samples, or (ii) distant fraudulent samples not belonging to any cluster where all clusters contain genuine samples, in which case it is treated as an outlier detection problem. The supervised learning approach takes known class labels and constructs a binary classifier. Fraud (including cyber fraud) detection is increasingly threatening and fraudsters always seem to be several steps ahead of organizations in finding new loopholes in the system and bypassing them effortlessly. On the other hand, organizations invest huge amounts of money, time, and resources to predict and mitigate fraud in the near, if not in real time. Financial fraud occurs in various areas such as banking, insurance, and investment (stock markets). It can be both offline and online. Offline frauds include accounting fraud, counterfeiting, and others.

Advances in technology and breakthroughs in deep learning models have increased intelligent automated trading and decision support systems in financial markets, especially equities and foreign exchange (FOREX). Fundamental analysis is using or mining text information such as financial news, company financial reports, and other economic factors such as government policies to predict price moves.

Table 3: Domains of application for smart data

Applications Domains	Limitations
Financial services	Fraud detection and prevention
Retail	Allowing brands to: - Analyzing customer sentiment - Offering personalized and contextual promotions
Telecommunications industry	Possibility of - Better allocating bandwidth based on real-time needs, and - diagnosing antenna condition
Understand the market conditions	Costs
Increased productivity	Security and Privacy Concerns
Fraud detection	/
Production	Preventive maintenance
Healthcare	- Monitoring patient vital signs, - Reducing readmission rates
Oil industry	- Proactive repairing infrastructures, and - Balancing the power delivered according to consumption
Public sector	- Detecting and preventing intrusion attempts on the network, and - Predicting epidemic risk
Transport sector	Possibility of detecting risky conduits

Big data is first and foremost data that is too large, too complex, too disparate, or too rapidly changing for correct and rapid analysis and use by current data processing methods. Smart data goes beyond this simple concept of mass data and refers to the useful and high-quality information that can be extracted from mass heterogeneous data, taking account of the quality, security, and protection of the data and how they are used, as well as the technical control of the mass of data. Knowledge is thus generated from the data. This data can then be used to generate

³<https://www.aps.dz/en/economy/39561-customs-over-385-000-counterfeit-goods-seized-in-2020>

value-added knowledge, underpinning this new data economy. So it's important to turn big data into smart data. Big data, which can be structured, unstructured, or semi-structured, enables rapid collection, storage, distribution, management, and analysis of large, structured data sets. Data from different sources may arrive at the system at different speeds. Thanks to parallelism, these large amounts of data can be processed, and thanks to Hadoop, big data can be structured and analyzed. Hadoop, open-source software for distributed processing of large datasets, is designed to scale from a single server to multiple machines with high fault tolerance. Smart Data always combines analysis of this data with human intelligence to provide comprehensive, synthesized, and useful business information (see Table 5).

. This model is intended to support an introductory crisis warning internship for citizens, interns, and future leaders. The training tool is used in three modes [2]. Novice mode allows the trainee to use extensive automated design and learning tools, such as observing and experimenting with different programs. It allows the learner to progressively learn by experiencing, observing, and failing. The novice mode allows him, at any point, to ask a smart education tool to generate the next step.

This tool analyses knowledge of this step, providing the optimal option and a list of relevant steps. The trainee can select any operation if he is not satisfied with the optimal operation suggested by the system. Online Manual Review mode allows the student to access all previous courses at any time by reviewing any previously learned concept, restarting any previously learned example, and resuming any case. This mode provides reference access to all learning, supporting example-based online help. Educational messaging helps raise situational awareness during public economic crises.

Many academic institutions are collaborating with hospitals, professional associations, and government and non-government organizations to design and adapt educational programs that promote practical training to prepare health workers and health systems to respond to the health needs of populations. The aim is the development of essential knowledge and skills so that best practice can be efficiently standardized in the management of new crises.

3. New Multimodal Deep Learning Model for Economic Environment Monitoring

The new multimodal CNN-LSTM model is presented. The LSTM layer handles temporal correlations (see Figure 2). Its enhancement with convolutional structures in both input and output states will solve the problem. By stacking several CNN-LSTM layers and building a coding prediction structure, the research created a network model for these spatiotemporal prediction problems (see Figure 3). The goal of crisis prediction is to use the previously observed sequence of social networks to prevent an event in a local area, such as Algiers, London, or Paris. From an automatic learning perspective, this is a spatiotemporal prediction problem. Assume a dynamic system represented by a $M \times N$ grid. Each cell of the grid contains P time-varying measures (word, bias). Each observation can be represented by a

tensor X of $R^P \times M \times N$, where R denotes the domain of observed features. If periodic observations are recorded, the study has a sequence of tensors $X_1, X_2, \dots, X_t$. This is formulated by Eq. (1) as the likeliest series of length K from all previous series of length J , including the current series:

$$\hat{Y}_{t+1}, \dots, \hat{Y}_{t+K} = \arg \max_{\mathbf{X}_{t+1}, \dots, \mathbf{X}_{t+K}} \left\{ p(\mathbf{X}_{t+1}, \dots, \mathbf{X}_{t+K} \mid \mathbf{Y}_{t-J+1}, \dots, \mathbf{Y}_{t-J+K}) \right\} \quad (1)$$

At each timestamp, a 2D map is observed. By dividing this map into non-tiled, non-overlapping patches and representing pixels within a patch as its measurements (using the CNN-LSTM multimodal deep learning model), the problem is naturally a spatiotemporal sequence prediction. This spatiotemporal sequence prediction problem differs from one-step time series prediction because this prediction target contains both spatial and temporal structures.

Content e , denoting input to the network, is defined by equation (2):

$$e = (w_1, \dots, w_1, \dots, w_n) \quad (2)$$

containing words $w_i \in W$, each from a finite vocabulary V . C_n is the set of social media content.

Once an error function is defined, the (learning) problem becomes one of optimizing: find the coefficient vector $\mathbf{w}^*$ that minimizes the error. In logistic regression, the error function is convex. A classifier is available once the optimal $\mathbf{w}^*$ coefficient vector is determined. This requires an independent test set to estimate the error probability of the classifier. CNNs are normalized variants of multilayer perceptron (each neuron is connected to the next layer). Full connectivity makes them susceptible to overfitting, as in (3).

$$\forall n \in [1, 2n_c^{[l]}] \text{Conv}(a^{[l-1]}, K_{x,y}^{(n)}) = \phi^{[l]} \left(\sum_{i=1}^{n_H^{[l-1]}} \sum_{j=1}^{n_W^{[l-1]}} \sum_{k=1}^{n_C^{[l-1]}} K_{i,j,k}^{(a)} * a_{x+i-1, y+j-1, k}^{[l-1]} * b_n^{[l]} \right) \quad (3)$$

$$\text{Dim}(\text{Conv}(a^{[l-1]}, K^{(n)})) = (n_H^{[l]}, n_W^{[l]})$$

CNNs use very little preprocessing: they learn filters designed by hand. The formula for the current state is as follows

$$h_s = f(h_{s-1}, x_s) \quad (4)$$

Applying activation function $\tanh$:

$$h_s = \tanh(\sigma_{hh} * h_{s-1} + \sigma_{xh} * x_s) \quad (5)$$

where

- $*$ denoting multiplication by elements.
- ω the function of the logistic sigmoid.
- i, f , and j are, respectively, input, forget, and output gates
- c the vector of cellular activation, the same size as hidden vector h , at level k .

The formula of current state follows as:

$$h_s = f(h_{s-1}, x_s) \quad (6)$$

applying the function of activation tanh:

$$h_s = \tanh(\sigma_{hh} * h_{s-1} + \sigma_{xh} * x_s) \quad (7)$$

where

- σ is weight.
- h denotes a sole vector that is hidden,
- σ_{hh} weight of the last hidden state,
- σ_{xh} weight of current input state,
- $\tanh$ the function of activation introducing an activation of non-linearity squashing to $[-1, 1]$.

Denoting output:

$$y_s = \sigma_{hy} * h_s \quad (8)$$

where

- y_s output state, and
- σ_{hy} weight of output state.

Consider a hidden-layer CNN-LSTM-based multimodal deep learning model with input content formulated by the equation:

$$e = (w_1, \dots, w_i, \dots, w_n)$$

containing words W , each from a finite vocabulary V , i.e. the set of texts issued by the social media, and giving as output the relevant text e_k . Let the following formula:

$$\forall i \in [1, N] \ e_i \in C^n = E \ \& \ e_i = (w_{i1}, \dots, w_{in})$$

containing words from the word set W . Each word built from a finite vocabulary V , the inclusion of the content of the source message is relevant to at least one tag or hash:

$$\exists j \in [1, M] \text{ such as } h_j \in H$$

It wants to learn a generic space as formulated by this formula:

$$E_K = \max_{k \in [1, K]} \left\{ e_k \right\}$$

normalize the differences in equation (4):

$$E_K = [E - RDF] \text{ where } RDF = [R + D + F] \quad (9)$$

where

- $R, D \ \& \ F$ denote respectively the set of duplicate re-tweets, duplicate contents and false alerts.
- Figure 1 shows how the multimodal hybrid model based on Deep CNN-LSTM works.

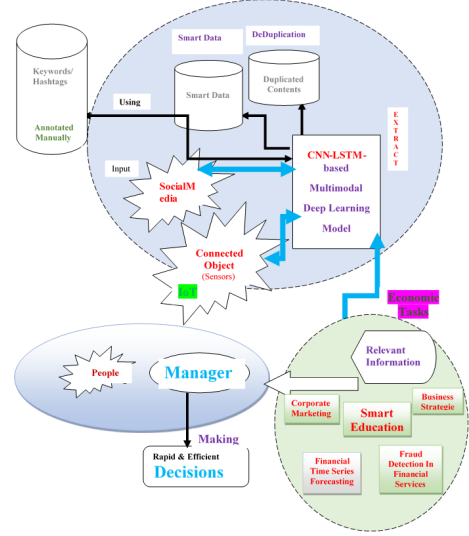


Figure 1: Functioning of the multimodal hybrid model based on Deep CNN-LSTM.

- The CNN-LSTM-based multimodal deep learning model can explain the transformation of e_i into e_k by

$$\exists j \in [1, M] \mid h_j \in H \ \& \ \exists l \in [1, L] \mid w_l \in W$$

$$\max_{k \in [1, K]} \left\{ e_i \rightarrow e_k = e_i \mid e_i \text{ is relevant for } h_j \ \& \ w_l \right\}$$

$$\text{with } i \in [1, N] \text{ and } e_i \notin [R + D + F]$$

Where

- $R, D \ \& \ F$ denote the duplicate retweets, duplicate content, and false alert set respectively.

The goal is then to maximize the size of the set K of E_K . Figure 3 shows the architecture of the multimodal Deep CNN-LSTM-based economic monitoring model.

Several tools for statistics and artificial intelligence are used with large data analysis techniques and recently Data Smart Streaming, the process of discovering new knowledge by analyzing large amounts of data in streaming, using deep learning, statistical, and mathematical techniques.

In the online tutorial mode, at any time during the training, the trainee has a menu to access all previous courses, such as:

- Present any previously learned concept,
- Demonstrate all examples and analyze problems explained or solved. This mode gives access to the course material as a reference [2, 3, 8]: thereby
- supporting online help based on examples. Training in commercial, financial, or fraud prediction [3] consists of:
- Advice to always pay attention to the quality of the products to be consumed, with the reinforcement of preventive hygiene, namely

Table 4: Comparing techniques and methods in models with education

Models	Educational Methods
[11]	An abstractive-extractive approach to situational tweet summarisation in crisis events
[3, 4, 5, 6, 7, 16]	Managing crises through education, based on CNN-LSTM
[13]	The importance of education during crises
[12]	Opportunities and challenges of education programs
[2]	FFNN-based model with smart education
[21]	RNN-based ALE to enhance awareness
[8]	LSTM-based model with smart education
New Approach	Traffic environment monitoring for warning, awareness and sustainable with safety education

- Check the quality, price, manufacture, and expiry date of each product
- Strengthening preventive hygiene,
- check that each product has indications visible on its packaging, how store it, manufacturing date, expiry date, batch number

- respect the conditions for using the product,
- Do not use anything without reading the instructions,
- End anarchic and uncontrolled buying,
- Promoting health and hygiene and
- Avoid inappropriate use (social, cultural).

This multimodal deep learning model is designed to support an introductory course aimed at preparing economic and business personnel, including managers, to meet population consumption needs. It consists of standardizing good service practices by developing essential skills, know-how, and knowledge in the field of expertise of companies and economic workers in general in the event of a crisis, namely:

- Skills appropriate to the sales service position or function on demand,
- Unique skills with a focus on their level, not the role,
- Specific roles and skills,
- Graded skills required according to managing the local economic process,
- Skills based on different target groups, and
- Application of transversal skills to business staff.

Information security is based on confidentiality, integrity, availability, non-repudiation (of a multiparty transaction) and authentication. Unfortunately, social media does not provide this information security. Information security is the main business and financial concept in social media. Here it is ensured in

two consecutive ways. First, security is against content retrieval from a multimodal deep learning approach with connected objects, smart data, and multiple social media sources [1, 2, 3, 8]. Then there is security against misinformation [3] (see table 10).

Table 5: Comparison of methods and techniques in fake news prevention models.

Models	Methods
[6]	Model economic environment monitoring to enhance economic activities in sustainability using deep learning with economic education from various sources
[7]	Smart education to manage emergency crises with secure information
[5]	Modelling CNN-LSTM-based model for economic monitoring and crisis forecasting in sustainability
[3]	Enhancing trade, business, financial time series prediction, fraud detection and marketing with DL, big data, HPC with safe information
[4]	Smart education with DL and social networks to manage sustainability with safe news
[16]	Fake news detection based on context
New Approach	CNN-LSTM-based Multimodal DL with Big Data for economic environment monitoring to improve Marketing, Business Strategies, Fraud Detection, Financial Time Series Forecasting

Table 9 shows the comparison between methods and techniques in models to Sustainability.

Table 6: Comparison of sustainable methods and techniques.

Models	Methods
[6]	Using deep learning with economic education from various sources to model the monitoring of the economic environment to strengthen economic activities in sustainability
[7]	Emergency crisis management with secure information through education
[5]	Modeling CNN-LSTM-based model for economic monitoring and crisis forecasting with safe news in sustainability
New Approach	CNN-LSTM-based Multimodal DL with Big Data for economic environment monitoring to improve Marketing, Business Strategies, Fraud Detection, Financial Time Series Forecasting

4. Results and Discussion

Here, the effectiveness of the proposed deep CNN-LSTM model was tested against the benchmark: the RNN and LSTMs prediction methods are the widely used deep learning models. In the experiment, these deep learning/machine learning models need to learn (find the best over-parameters), including finding the number of neurons, the number of layers of neural nets, and the activation function of the deep CNN-LSTM multimodal deep learning model. In addition to this, it was also possible to determine final configuration values. The research aims to investigate global changes that affect the vulnerability of populations to climate change and extreme events and to help develop strategies to help communities better cope with the effects of climate change.

Electricity fraud detection approach [14] using a combination of Long Short-Term Memory (LSTM) methods and robust Exponential and Holt-Winters Smoothing (EHWS) methods, using sensors to improve the accuracy and efficiency of fraud detection mechanisms, while studying the fraudulent behavior of electricity consumers. It addresses the huge problem of non-technical losses (NTL), also known as electricity fraud, for electricity distribution utilities.

Table 7: Examples of relevant content of electricity fraud detection in Algeria

Models	Electricity fraud detection in Algeria
Naive Bayes (NB) [26]	2233
Random Forest (RF) [27]	2111
Support Vector Machine (SVM) [17]	2444
Neural Network (NN) via Twitter & Facebook [1]	2812
Feed-forward Neural Network (FFNN) via All the Web [2]	3161
Recurrent Neural Network (RNN) via Smart Data & All the Web [3]	2515
LSTM via Smart Data & All the Web [3]	2924
Hybrid of Deep CNN-LSTM via All the Web [17, 18]	2664
New Approach via Sensor, Smart Data & All the Web	6118

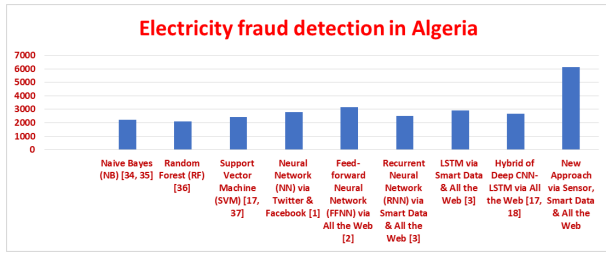


Figure 2: Examples of Relevant Content of electricity fraud detection in Algeria from March 2020 to date.

A total of 385,714 fake goods will be confiscated by the Algerian tax authorities in 2020, according to the anti-fraud department of the Algerian tax administration ⁴.

The goal is maximizing the size K of the set E_K . Reliable connectivity and data security enable seamless, efficient services for remotely controlling and monitoring fire alarms. Remote services turn the device into a state-of-the-art IoT solution. It allows remote access to fire detection tools to manipulate, maintain, and live monitor the contamination level of fire detectors and alarms, and to report faults to intelligent devices.

Table 12 is an example COVID-19 assessment. Examples of relevant content for a set of hashtags and keywords from multiple sources for the Algerian forest fire of August 16th – 19th, 2021. Each message is represented by a sequence of transactions $T = (t_1, \dots, t_n)$. COVID-19 for Algeria, from March 2020

⁴<https://www.aps.dz/en/economy/39561-customs-over-385-000-counterfeit-goods-seized-in-2020>

until today, for a set of hashtags and keywords for all social networks.

5. Conclusion and Perspectives

In conclusion, using a CNN-LSTM-based multimodal deep learning model for economic monitoring and crisis forecasting can fundamentally change how people, especially managers, improve economic activities. The study investigated integrating IoT, big data, HPC, and multiple social media technologies for economic monitoring and crisis forecasting. The results provide insight into the perceived impact on warning, effectiveness, user satisfaction, and favorable attitudes. It shows that multimodal deep learning based on CNN-LSTM can improve economic activities and crisis prediction. Citizens and administrators may conduct economic transactions in full confidence, in a regulated and secure environment where information will be protected. The results of the research agree with the idea that this new multimodal deep learning approach for economic monitoring and crisis forecasting systems can bridge the gap between theoretical understanding and practical implementation.

Although the research highlights the possible benefits of this new approach, the validation of this new system will require feedback on the use of this new approach for economic surveillance and crisis prediction using descriptive statistics through sampling. This will be the role of future work. The study presented an ad hoc real-time multimodal deep learning model. It is based on a CNN-LSTM for educational warning to prevent events. These include data from networked objects, smart data, and multiple social media. Such an approach is particularly useful for improving economic activity. The community can be alerted to potential events, informed of status, and request help. With data, the content is written informally, with no syntax, logic, or noise. There are also indicators of pollution, temperature, humidity, etc. Therefore, it may be biased towards certain domains. Multimodal deep learning model features are developed by analyzing specific event content.

In future works, this research study has several potential applications. Validating relevant information (avoiding misuse), using different languages (including Arabic and French), and extracting the most useful to allow the community to live easily and comfortably, saving the lives of those stuck under the rubble, avoiding any complications, difficulties, material, social or even professional. The real-time paradigm could be extended by using big data to search for information about past events, validating a possible warning to keep people safe. In addition, quantitative research involves surveying a sample of clients and managers to gauge their views on how to improve economic monitoring and crisis forecasting through education.

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Declarations

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